EE2021 -

Materials Science for Electrical Engineers

Mechanical Properties

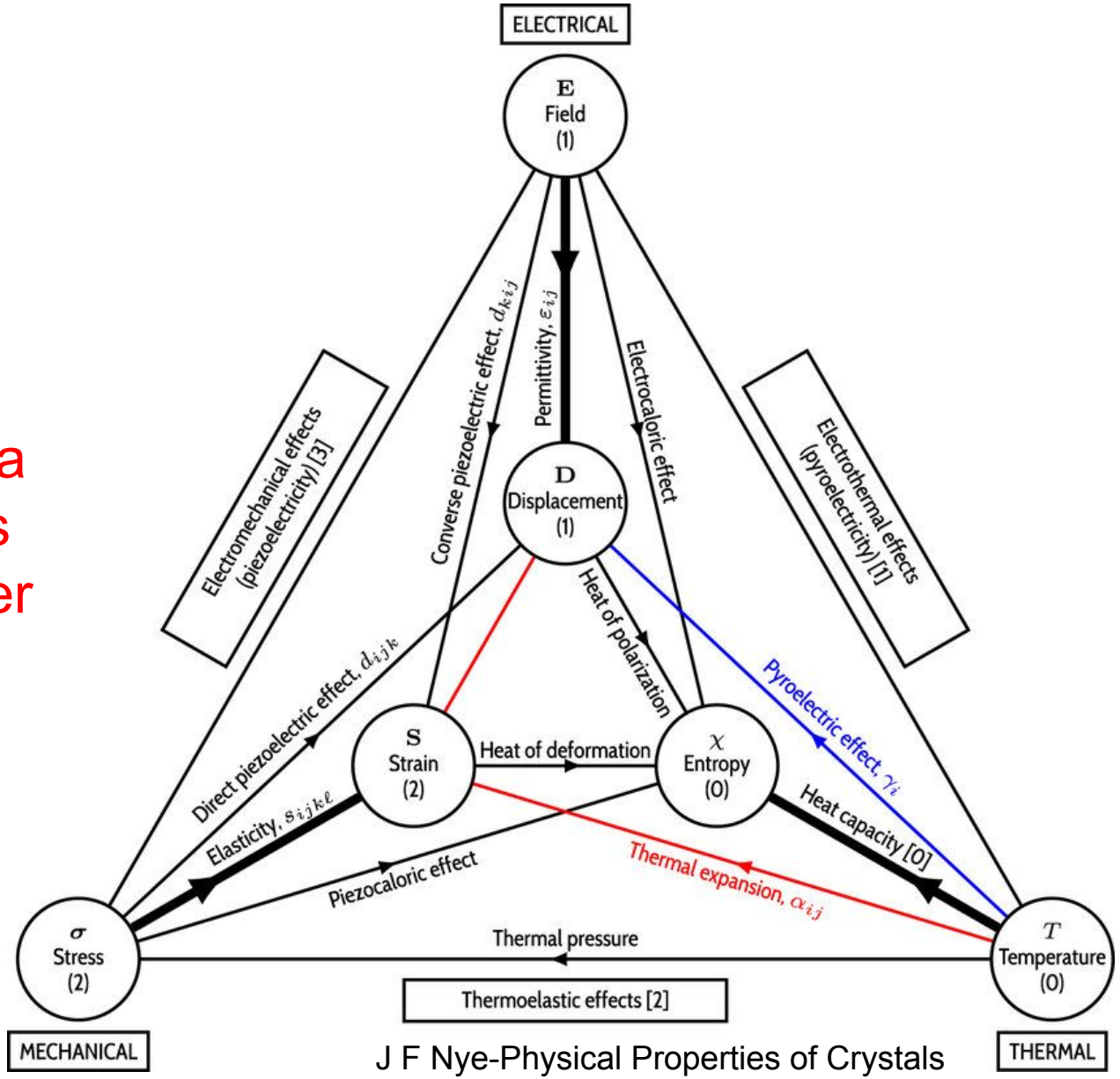
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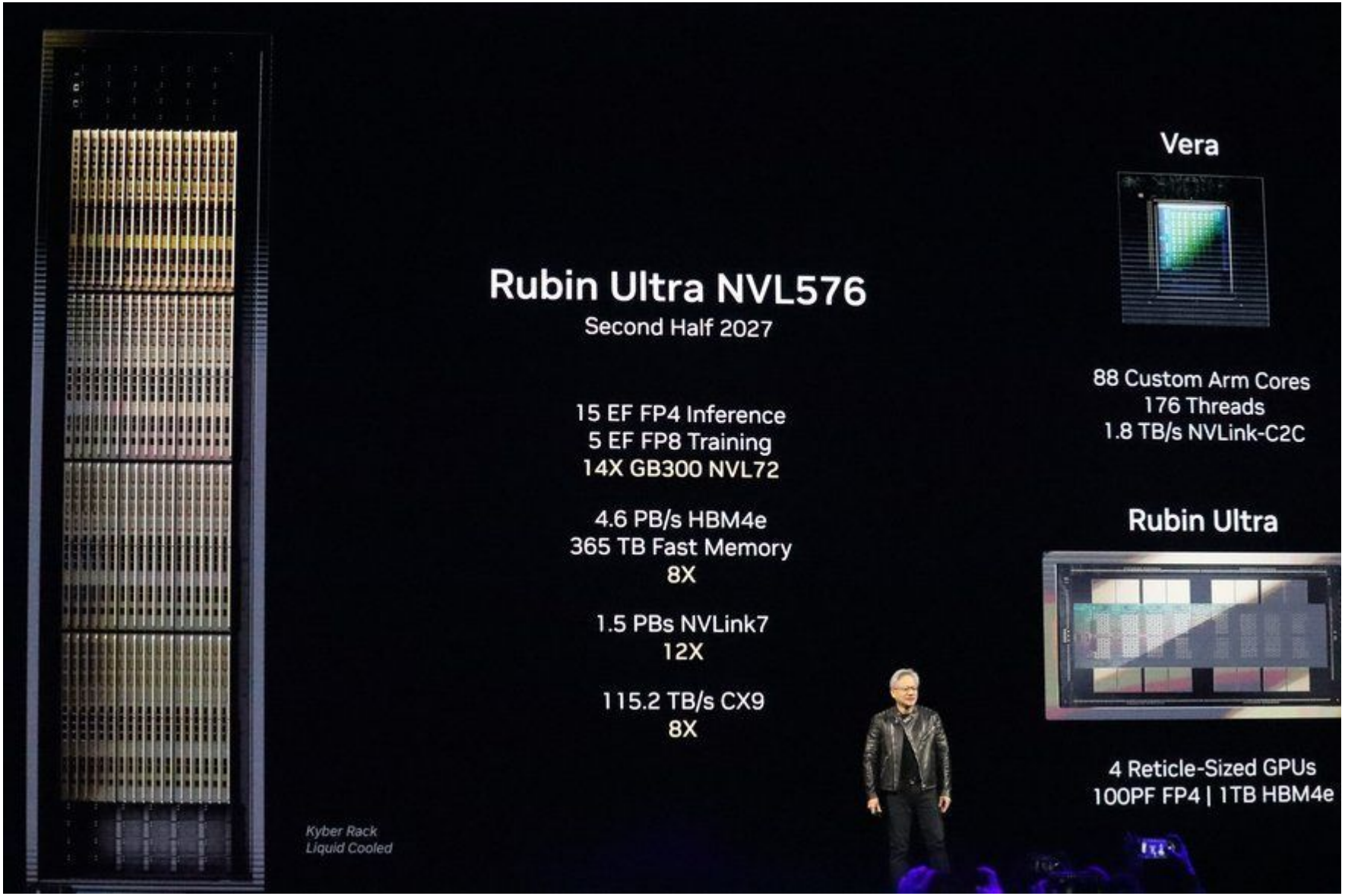
Why relationship between Electrical, Mechanical and Thermal properties are important

Quality - Product meets design specifications and defect free

Reliability - Ability of a product to perform its intended function over time without any failure.



J F Nye-Physical Properties of Crystals



"Each rack is 600kW, 2.5 million parts, and obviously a whole lot of GPUs," CEO Huang said

https://www.tomshardware.com/pc-components/gpus/nvidia-shows-off-rubin-ultra-with-600-000-watt-kyber-racks-and-infrastructure-coming-in-2027

Stress and Strain

Tensor vs Matrix

- Anisotropy- physical property of a crystal depends on the direction in which it is measured
- All crystal are anisotropic for some of their properties
- Tensors of different ranks are used to represent such properties
- A tensor of rank r in 3-D space has 3^r components. Each component has r suffixes
- The components of a rank 1 tensor (vector) change under a rotation of axes as, $X'_i = R_{ij}X_j$, where R is the 3×3 matrix of direction cosines

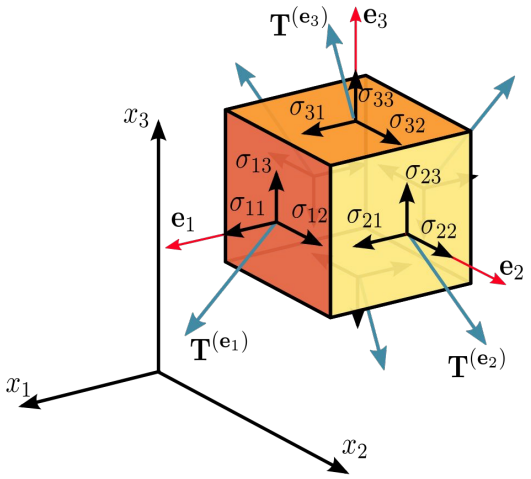
$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- The transformation law for a tensor of rank 2 is $T'_{ij} = R_{ik}R_{jl}T_{kl}$ or in matrix notation $T'=RTR^T$
- A second rank Tensor component T_{ij} transforms like the product x_ix_j
- A r rank Tensor component transforms like the product $x_1x_2\ldots x_r$
- Tensor may be represented by a matrix, but not all matrices are tensors. Matrix is mathematical idea. Any set of numbers can be arranged in a rectangular array. A tensor is a physical concept
- Tensor in ML/DS/AI is another name for a multidimensional array.

J F Nye-Physical Properties of Crystals

Stress

- Consider a unit cube within the body. The force transmitted across each face are resolved into three components
- σ_{12} is the stress on the face normal to x_1 direction in the $+x_2$ direction by the material outside the cube upon the material inside it.
- If the stress is homogeneous, the forces exerted on the cube on the opposite faces must be equal and opposite
- σ_{11}, σ_{22}, σ_{33} are normal and σ_{12}, σ_{21}, σ_{31} , σ_{13}, σ_{23}, σ_{32} are shear components
- For pressure, the force acting on the plane is always normal to the plane, whereas with stress the force can be inclined to the plane normal and even in the plane.



Tensile stress - positive
Compressive stress - negative

For static equilibrium with no torque, $\sigma_{ij} = \sigma_{ji}$

$$\begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$

Stress tensor (rank 2) can be written as

Types of Stress

Uniaxial

$$\begin{bmatrix} \sigma & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Biaxial

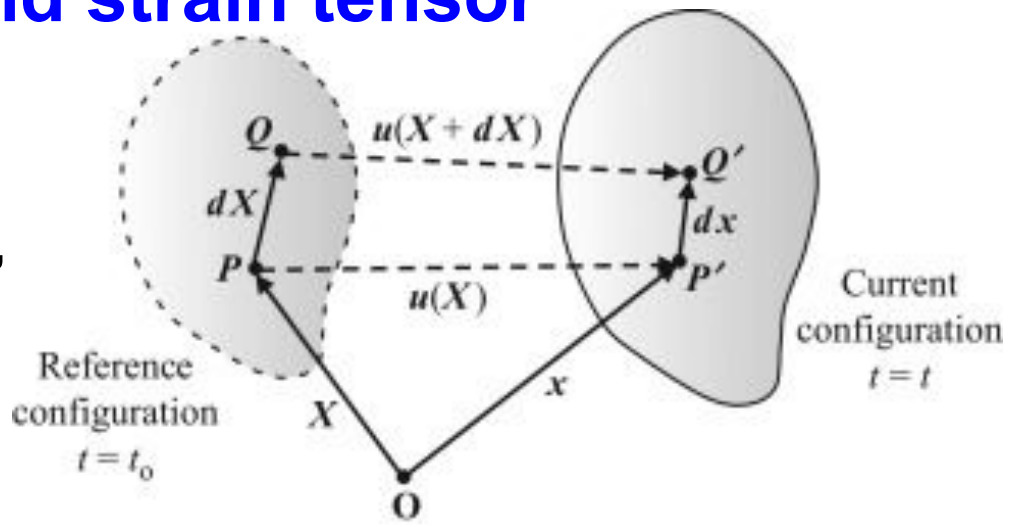
$$\begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hydrostatic pressure

$$\begin{bmatrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{bmatrix}$$

Displacement gradient and strain tensor

Suppose X and X+dX become x and x+dx in the deformed state. We may express x(X) as X+u(X), where u(X) is the displacement undergone by a point at X in the undeformed state



The displacement gradient tensor that describes the local variation of the displacement field within a material can be written as sum of symmetric and anti-symmetric part

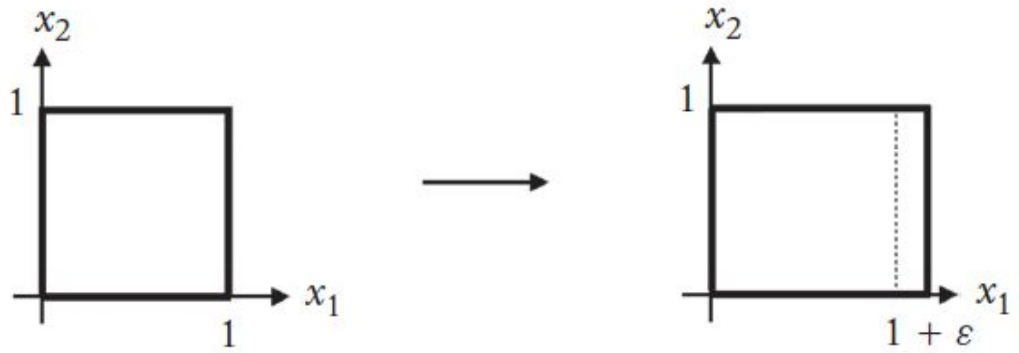
$$\begin{bmatrix} \frac{\partial u_x}{\partial x} & \frac{\partial u_x}{\partial y} & \frac{\partial u_x}{\partial z} \\ \frac{\partial u_y}{\partial x} & \frac{\partial u_y}{\partial y} & \frac{\partial u_y}{\partial z} \\ \frac{\partial u_z}{\partial x} & \frac{\partial u_z}{\partial y} & \frac{\partial u_z}{\partial z} \end{bmatrix} = \varepsilon_{kj} + \omega_{kj}$$

$$\varepsilon_{kj} = \frac{1}{2}\left(\frac{\partial u_k}{\partial X_j} + \frac{\partial u_j}{\partial X_k}\right) \qquad \omega_{kj} = \frac{1}{2}\left(\frac{\partial u_k}{\partial X_j} - \frac{\partial u_j}{\partial X_k}\right)$$

Strain tensor

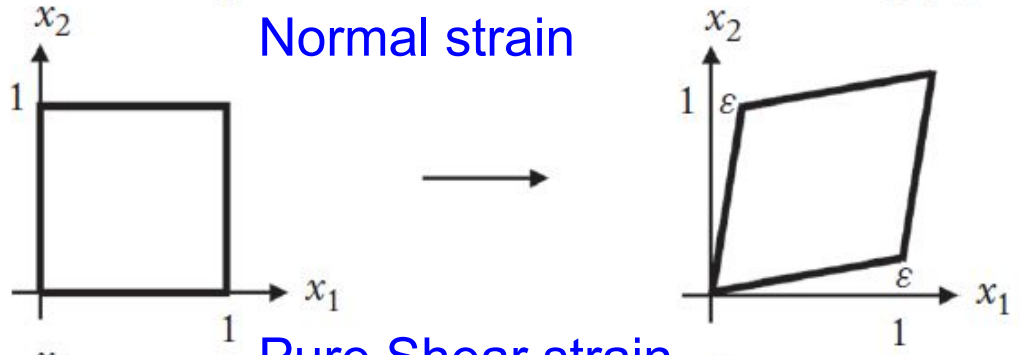
Rotation tensor

Types of Strain



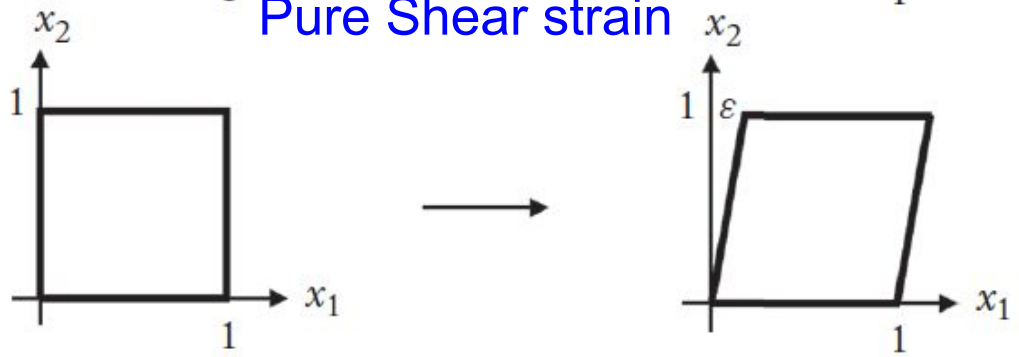
Normal strain

$$\mathbf{e} = \begin{pmatrix} \varepsilon & 0 \\ 0 & 0 \end{pmatrix}$$



Pure Shear strain

$$\mathbf{e} = \begin{pmatrix} 0 & \varepsilon \\ \varepsilon & 0 \end{pmatrix}$$



Simple shear

$$\begin{pmatrix} 0 & \varepsilon/2 \\ \varepsilon/2 & 0 \end{pmatrix} + \begin{pmatrix} 0 & \varepsilon/2 \\ -\varepsilon/2 & 0 \end{pmatrix}$$

pure shear strain of $\varepsilon/2$ and a rotation by $\theta = \varepsilon/2$

Does Stress cause Strain or vice versa?

Stress does not cause strain (nor does strain cause stress), but the two are coupled to each other, the presence of either necessarily implies that the other is also present.

Internal deformations for example, can be “caused” by thermal gradients, dislocations, and defects in the crystal lattice or by the presence of dopant atoms that are significantly larger or smaller than the host atoms and thus deform the lattice structure. In such cases, internal forces are established, and it would be proper to refer to these stresses as being the result of the strains.

A further distinction between stress and force comes from the fact that stresses always occur in opposite (but not always equal) pairs. In the static case, the stresses are equal because there is no net motion of any internal volumes. In the dynamic case (e.g., the propagation of an acoustic wave), the opposite stresses are not generally equal.

Generalized Hooke's law

- In linear elasticity theory (displacement gradient <<1), the strain induced in a medium is proportional to the applied stress and vice versa. The constants of proportionality can be expressed as a fourth rank tensor

$$\varepsilon_{lj}=S_{ijkl}\sigma_{kl} \qquad \sigma_{lj}=C_{ijkl}\varepsilon_{kl}$$

S_{ijkl} *is the elastic compliance tensor* C_{ijkl} *is the elastic stiffness tensor*

Though it appears at first sight that there are 3^4 = 81 independent elastic constants, crystal symmetry make it a smaller number. The smallest number of independent elastic constants is just two (elastically isotropic material like rubber). In cubic crystals (Si, Ge) there are just three independent elastic constants and in hexagonal crystals five. The largest number of independent elastic constants in any material is 21. This in the case in a triclinic crystal where there are no rotational symmetries in the point group.

Generalized Hooke's law for Cubic materials

For cubic crystals can be expressed in a convenient matrix form

$$\begin{bmatrix}\sigma_1\\ \sigma_2\\ \sigma_3\\ \sigma_4\\ \sigma_5\\ \sigma_6\end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{12} & 0 & 0 & 0\\ c_{12} & c_{11} & c_{12} & 0 & 0 & 0\\ c_{12} & c_{12} & c_{11} & 0 & 0 & 0\\ 0 & 0 & 0 & c_{44} & 0 & 0\\ 0 & 0 & 0 & 0 & c_{44} & 0\\ 0 & 0 & 0 & 0 & 0 & c_{44}\end{bmatrix} \begin{bmatrix}\varepsilon_1\\ \varepsilon_2\\ \varepsilon_3\\ \varepsilon_4\\ \varepsilon_5\\ \varepsilon_6\end{bmatrix}$$

$$S_{11} = \frac{C_{11}+C_{12}}{(C_{11}-C_{12})(C_{11}+2C_{12})}$$

$$S_{12} = \frac{-C_{12}}{(C_{11}-C_{12})(C_{11}+2C_{12})}$$

$$S_{44} = \frac{1}{C_{44}}.$$

SIMPLIFIED SUBSCRIPTS FOR CUBIC SYMMETRY

Face/Direction	Condensed Numeric Equivalent
xx or 11	1
yy or 22	2
zz or 33	3
yz or 23	4
zx or 31	5
xy or 12	6

$$\begin{bmatrix}\varepsilon_1\\ \varepsilon_2\\ \varepsilon_3\\ \varepsilon_4\\ \varepsilon_5\\ \varepsilon_6\end{bmatrix} = \begin{bmatrix} s_{11} & s_{12} & s_{12} & 0 & 0 & 0\\ s_{12} & s_{11} & s_{12} & 0 & 0 & 0\\ s_{12} & s_{12} & s_{11} & 0 & 0 & 0\\ 0 & 0 & 0 & s_{44} & 0 & 0\\ 0 & 0 & 0 & 0 & s_{44} & 0\\ 0 & 0 & 0 & 0 & 0 & s_{44}\end{bmatrix} \begin{bmatrix}\sigma_1\\ \sigma_2\\ \sigma_3\\ \sigma_4\\ \sigma_5\\ \sigma_6\end{bmatrix}$$

$$\begin{bmatrix}\varepsilon_{xx}\\ \varepsilon_{yy}\\ \varepsilon_{zz}\\ \varepsilon_{yz}\\ \varepsilon_{zx}\\ \varepsilon_{xy}\end{bmatrix} = \begin{bmatrix} \frac{1}{E_x} & -\frac{\nu_{yx}}{E_y} & -\frac{\nu_{zx}}{E_z} & 0 & 0 & 0\\ -\frac{\nu_{xy}}{E_x} & \frac{1}{E_y} & -\frac{\nu_{zy}}{E_z} & 0 & 0 & 0\\ -\frac{\nu_{xz}}{E_x} & -\frac{\nu_{yz}}{E_y} & \frac{1}{E_z} & 0 & 0 & 0\\ 0 & 0 & 0 & \frac{1}{G_{yz}} & 0 & 0\\ 0 & 0 & 0 & 0 & \frac{1}{G_{zx}} & 0\\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{xy}}\end{bmatrix} \begin{bmatrix}\sigma_{xx}\\ \sigma_{yy}\\ \sigma_{zz}\\ \sigma_{yz}\\ \sigma_{zx}\\ \sigma_{xy}\end{bmatrix}.$$

S_{ij} coefficients are converted to Young's modulus (E), Poisson's ratio (v), and the shear modulus (G)

M. A. Hopcroft, W. D. Nix and T. W. Kenny, "What is the Young's Modulus of Silicon?," JMEMS, April 2010

Generalized Hooke's law for Silicon

The Young's modulus, Poisson's ratio and Shear modulus of Silicon varies with crystal orientation. The matrix is given for two important directions.

$$\begin{bmatrix}\sigma_1\\ \sigma_2\\ \sigma_3\\ \sigma_4\\ \sigma_5\\ \sigma_6\end{bmatrix} = \begin{bmatrix}165.7 & 63.9 & 63.9 & 0 & 0 & 0\\ 63.9 & 165.7 & 63.9 & 0 & 0 & 0\\ 63.9 & 63.9 & 165.7 & 0 & 0 & 0\\ 0 & 0 & 0 & 79.6 & 0 & 0\\ 0 & 0 & 0 & 0 & 79.6 & 0\\ 0 & 0 & 0 & 0 & 0 & 79.6\end{bmatrix} \begin{bmatrix}\varepsilon_1\\ \varepsilon_2\\ \varepsilon_3\\ \varepsilon_4\\ \varepsilon_5\\ \varepsilon_6\end{bmatrix} \text{ GPa}$$

Cubic Silicon ([100], [010], [001]). (x,y,z directions)

$$\begin{bmatrix}\sigma_1\\ \sigma_2\\ \sigma_3\\ \sigma_4\\ \sigma_5\\ \sigma_6\end{bmatrix} = \begin{bmatrix}194.5 & 35.7 & 64.1 & 0 & 0 & 0\\ 35.7 & 194.5 & 64.1 & 0 & 0 & 0\\ 64.1 & 64.1 & 165.7 & 0 & 0 & 0\\ 0 & 0 & 0 & 79.6 & 0 & 0\\ 0 & 0 & 0 & 0 & 79.6 & 0\\ 0 & 0 & 0 & 0 & 0 & 50.9\end{bmatrix} \begin{bmatrix}\varepsilon_1\\ \varepsilon_2\\ \varepsilon_3\\ \varepsilon_4\\ \varepsilon_5\\ \varepsilon_6\end{bmatrix} \text{ GPa}$$

Standard (001) Silicon Wafer ([110], [$\bar{1}$10], [001]). (x,y,z directions)

Young's modulus (E), Poisson's ratio (v), and the shear modulus (G) in the axes of interest (x, y, z);

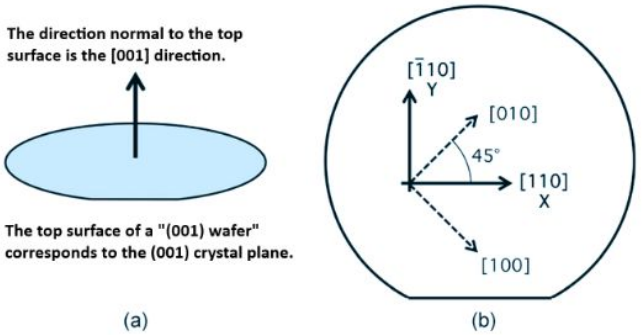


Fig. 2: Crystal orientation of a standard (001) silicon wafer. (a) The wafer surface is aligned with the (001) plane. (b) The in-plane X- and Y-axes are oriented along the [110] and [$\bar{1}$10] directions, respectively.

M. A. Hopcroft, W. D. Nix and T. W. Kenny, "What is the Young's Modulus of Silicon?," JMEMS, April 2010

Isotropic materials (eg: Tungsten)

$$\varepsilon_1 = \frac{\sigma_1}{Y} - \frac{\nu\sigma_2}{Y} - \frac{\nu\sigma_3}{Y} \qquad \varepsilon_4 = \frac{2(1+\nu)}{Y}\sigma_4$$

$$\varepsilon_2 = \frac{\sigma_2}{Y} - \frac{\nu\sigma_1}{Y} - \frac{\nu\sigma_3}{Y} \qquad \varepsilon_5 = \frac{2(1+\nu)}{Y}\sigma_5$$

$$\varepsilon_3 = \frac{\sigma_3}{Y} - \frac{\nu\sigma_1}{Y} - \frac{\nu\sigma_2}{Y} \qquad \varepsilon_6 = \frac{2(1+\nu)}{Y}\sigma_6$$

Consider a tensile test where a sample is loaded in tension along the x_3 axis and no constraints or loads are applied along x_1 and x_2 (σ_1=σ_2=0)

$$\nu = -\frac{\varepsilon_1}{\varepsilon_3} = -\frac{\varepsilon_2}{\varepsilon_3} \qquad Y = \frac{1}{S_{11}},\ \nu = -\frac{S_{12}}{S_{11}}. \qquad G = \frac{Y}{2(1+\nu)}$$

For isotropic materials, only two parameters are independent. Poisson's ratio($-1 \le \nu \le 1/2$) is the ratio of the contraction in the lateral x_1 and x_2 directions to the tensile strain along x_3.

Band structure engineering in semiconductors

Mechanical Stress

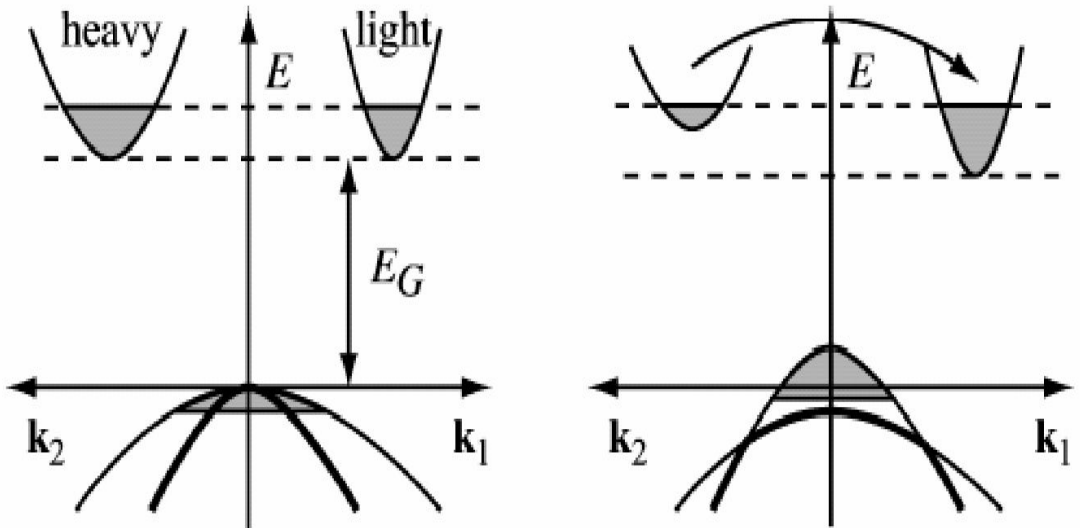
deforms lattice and changes band structure

Lifts the degeneracy of CB bottom and VB maximum, warps/shifts bands

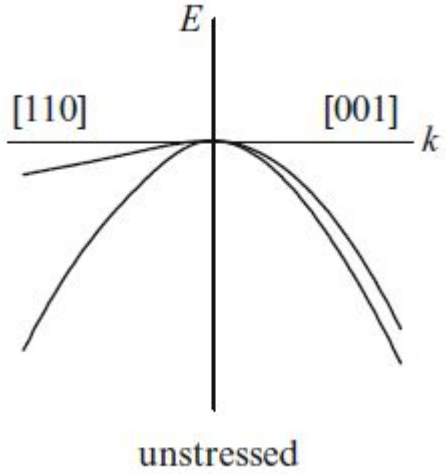
Puts more carriers in a low effective mass valley/large DOS

less intervalley scattering with fewer valleys

better mobility



Si valence bands under stress



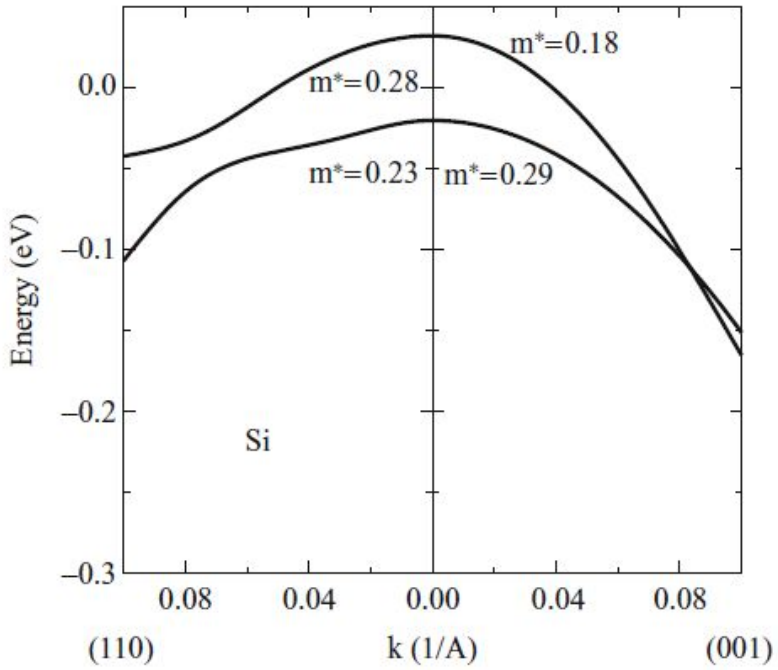


Fig. 4.31. Si valence bands under biaxial tensile stress

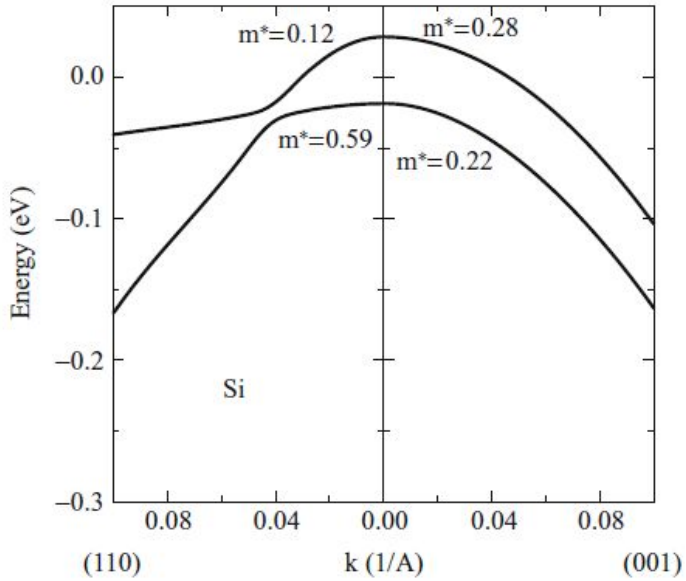
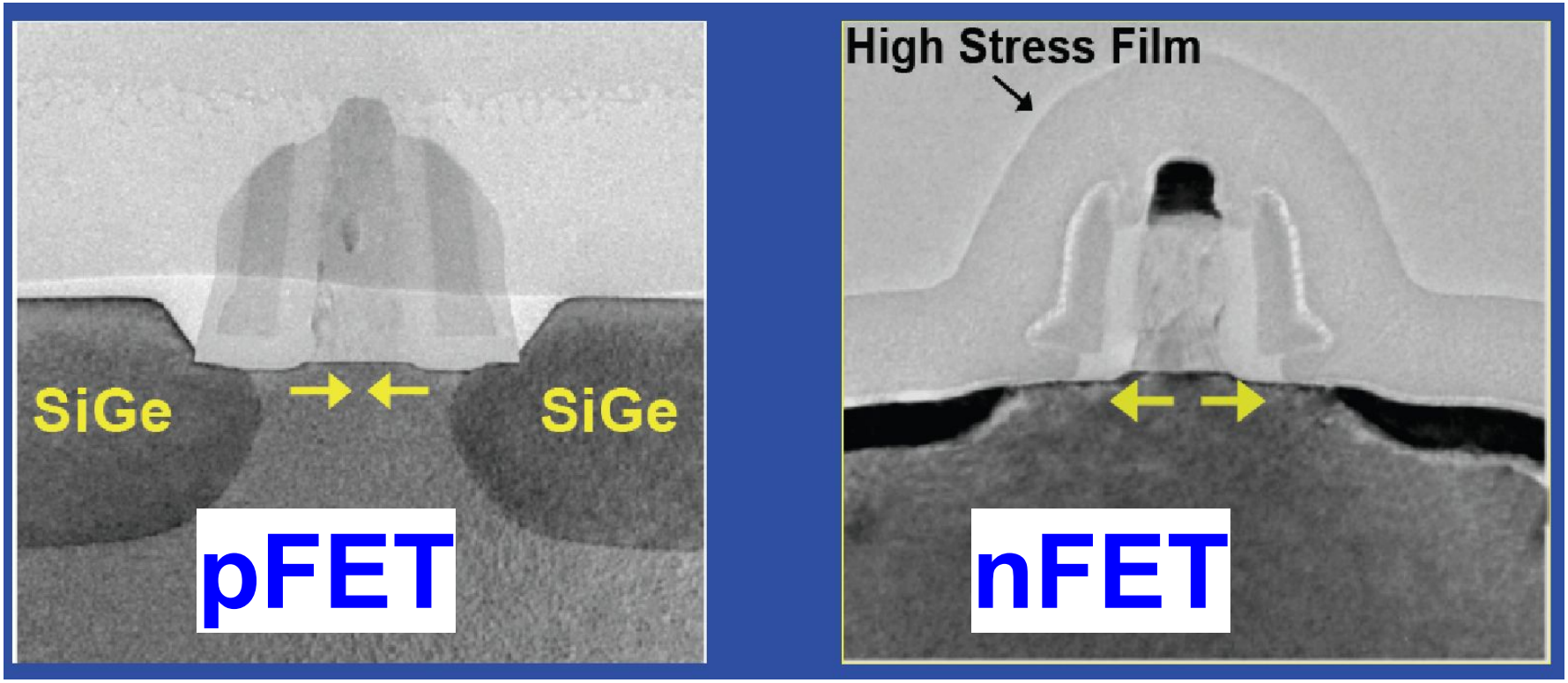


Fig. 4.32. Si valence bands under uniaxial compressive stress

Strained Silicon technology - Intel 90nm

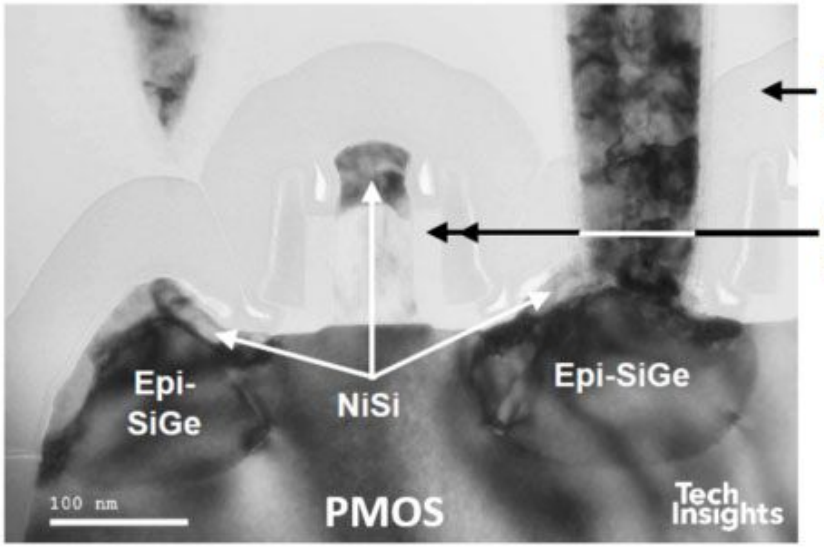
- PMOS - SiGe source/drain - Compressive uniaxial stress in direction current flow increases hole mobility.
  - 50% - 70% gain in drive current.
- NMOS - High stress SiN liner - Tensile uniaxial stress in direction of current flow increase electron mobility.
  - 10% - 15% gain in drive current.

K. Mistry et al, VLSI Tech. Sym. 2004

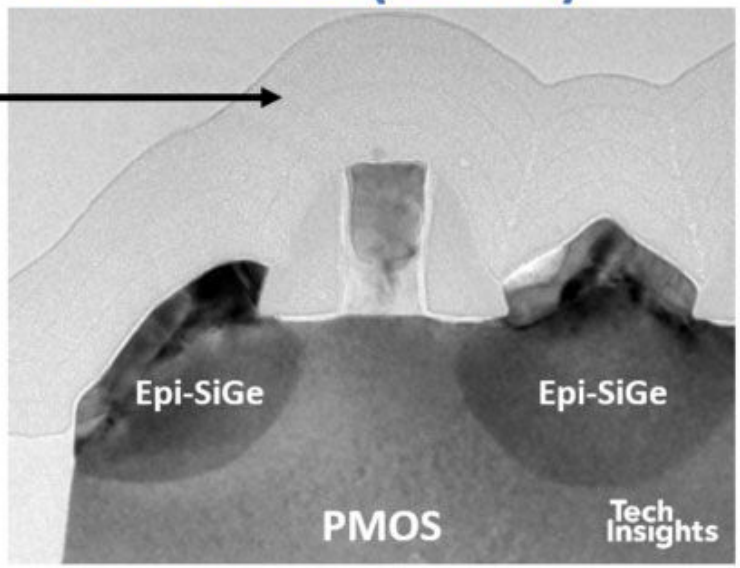


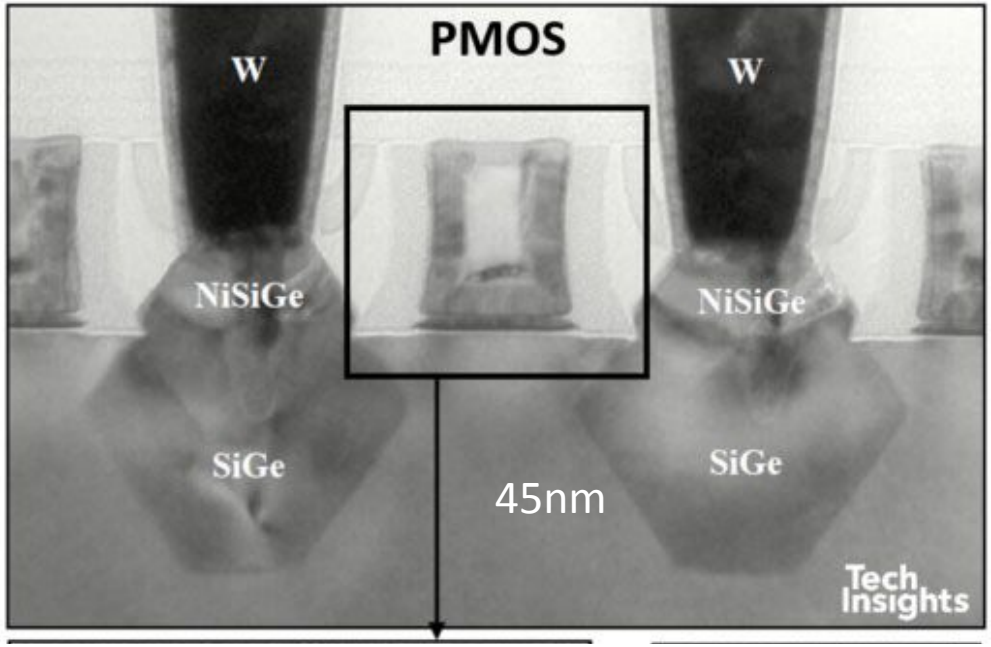
SiGe profile and molefaction engineering

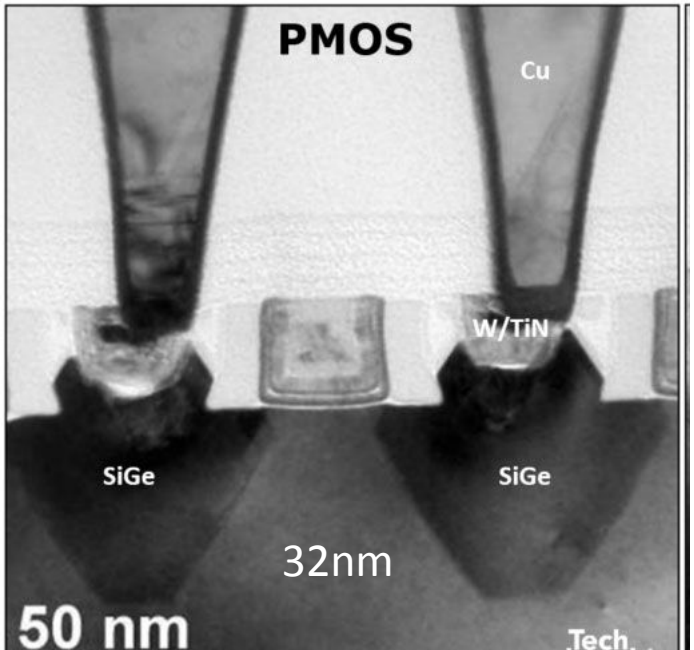
Intel Prescott (90 nm)



Intel Presler (65 nm)

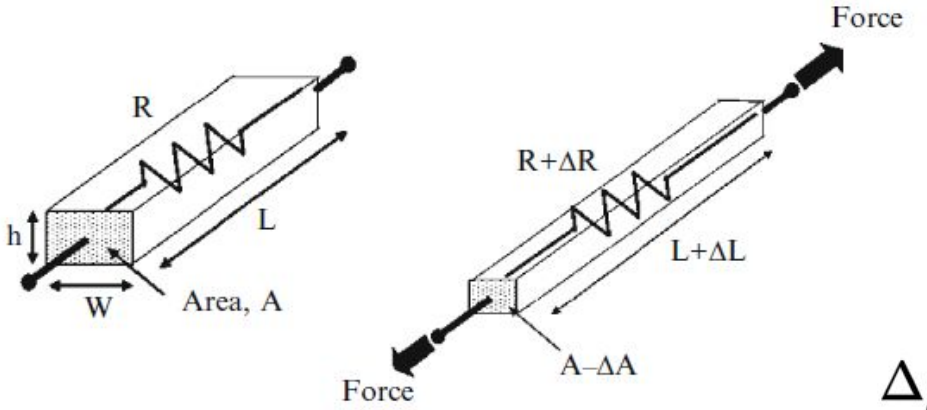






Piezoresistivity

Piezoresistivity is an effect of stress/strain-induced resistivity change of a material. The underlying physics is same as that of mobility enhancement in MOS transistors - modify bandstructure to change effective mass and scattering rates. While the strain in transistors is fixed (why?), piezoresistivity strain sensors are designed to transduce/detect varying strains and produce a proportional change in resistance.



$$\frac{\Delta R}{R} = \frac{\Delta L}{L} - \frac{\Delta W}{W} - \frac{\Delta H}{H} + \frac{\Delta \rho}{\rho}$$

$$\frac{\Delta R}{R} = (1 + 2\nu)\varepsilon_{\text{axial}} + \frac{\Delta \rho}{\rho}$$

$$\frac{\Delta \rho_{ij}}{\rho} = \sum_{k,j} \pi_{ijkl}\tau_{kl} \qquad \frac{\Delta \rho_i}{\rho} = \sum_{k=1}^{6} \pi_{ik}\tau_k$$

$$\pi_{ik} = \begin{pmatrix} \pi_{11} & \pi_{12} & \pi_{12} & 0 & 0 & 0 \\ \pi_{12} & \pi_{11} & \pi_{12} & 0 & 0 & 0 \\ \pi_{12} & \pi_{12} & \pi_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & \pi_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & \pi_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & \pi_{44} \end{pmatrix}$$

Piezoresistance tensor (rank 4) in matrix form (only 3 indpt entries for Si)

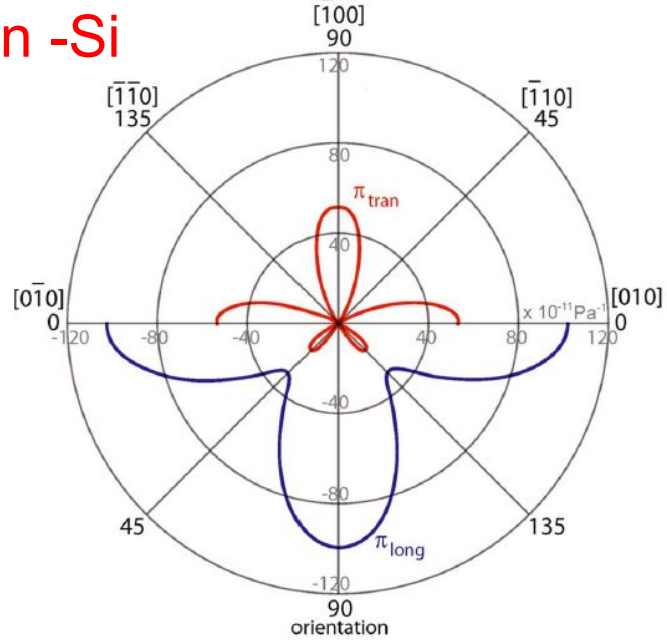
Yongke Sun et al, Strain effect in Semiconductors

Piezoresistivity

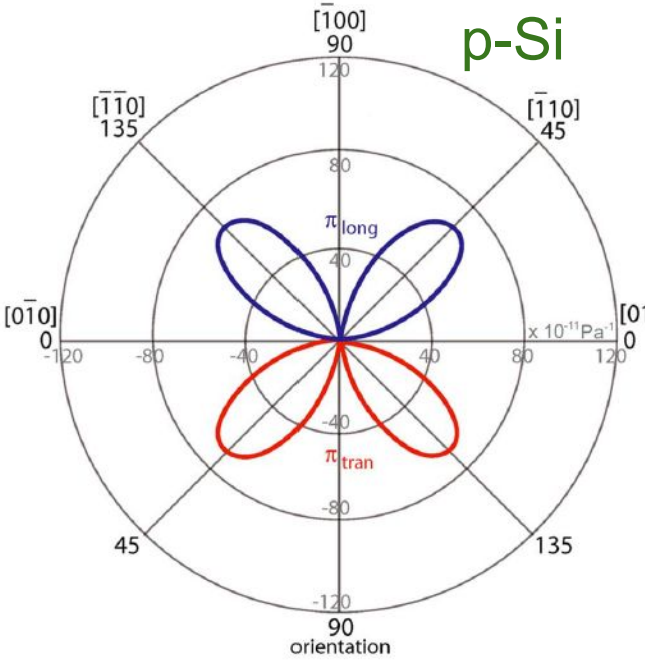
Consider the special case where the stress T_1 is longitudinal with respect to the current density, J_1, π_{11} may be considered as the longitudinal coefficient, π_L. The stress T_2 is transverse with respect to the current density, J_1, so π_{12} may be considered as the transverse coefficient, π_T.

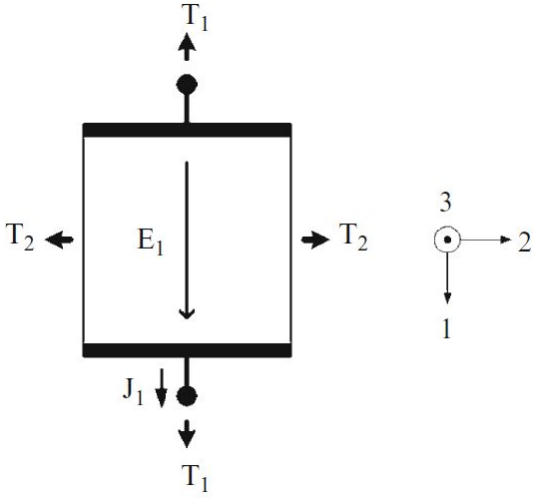
$$\frac{\Delta\rho_1}{\rho} = \pi_L T_1 + \pi_T T_2$$

n -Si



p-Si



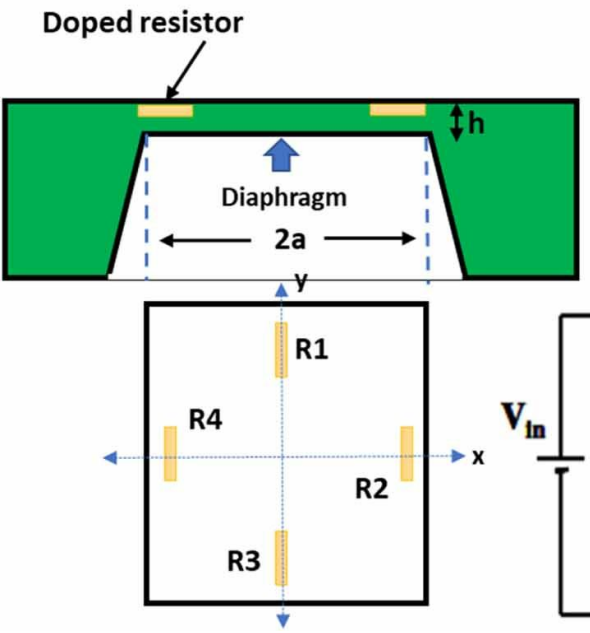


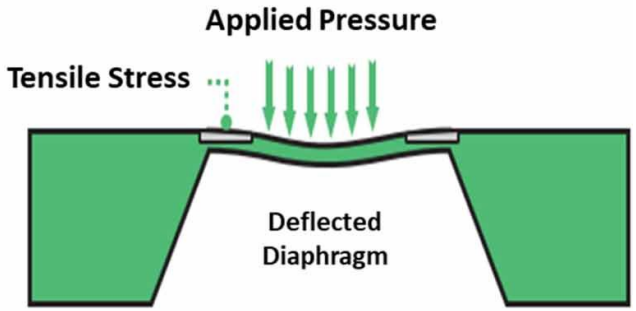
How much stress is needed to reduce the resistance ($\Delta\rho/\rho$) in <110> direction by 30% for n & p type?

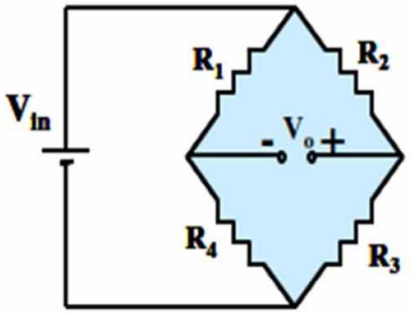
π_l and π_T values (in units of 10^{-11} Pa^{-1}) for (001) Si. Upper half means +ve and lower half means -ve values. Tensile stress - positive and Compressive stress - negative.

K. E. Petersen, "Silicon as a mechanical material," in *Proceedings of the IEEE*,, May 1982
A. A. Barlian et al, "Review: Semiconductor Piezoresistance for Microsystems," in *Proceedings of the IEEE*,, March 2009

Piezoresistive pressure sensor



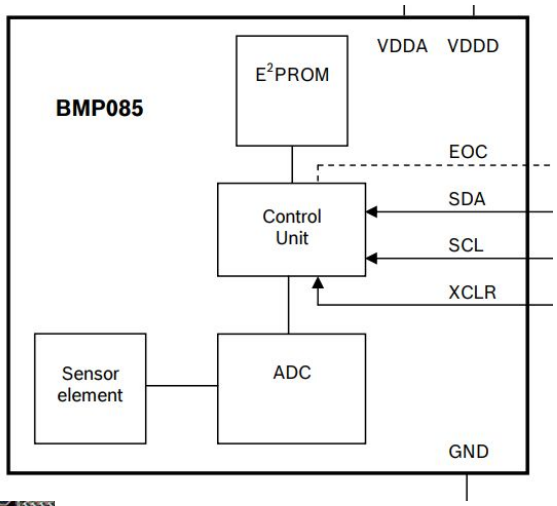


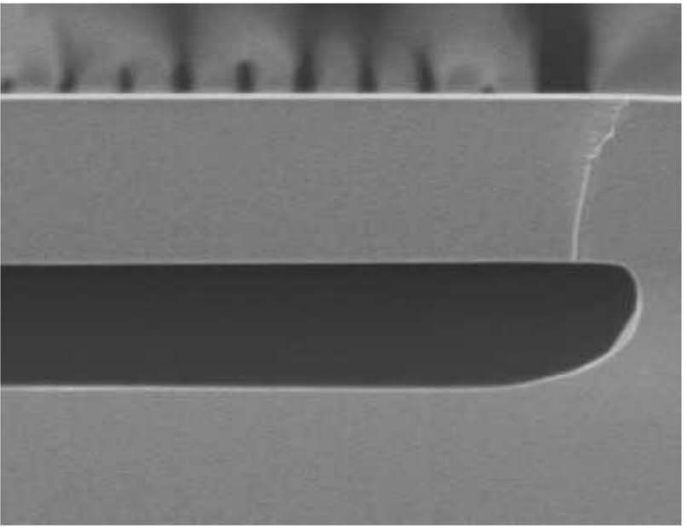


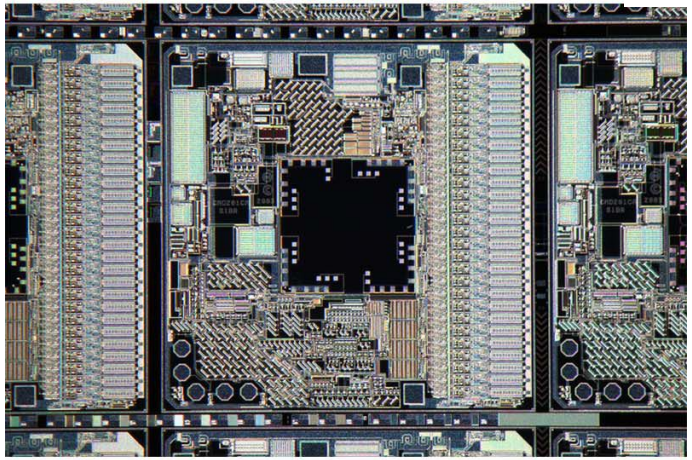
R1=R2=R3=R4=R

$$V_{OUT} = \frac{\Delta R}{R} V_{in}$$

$$\frac{\Delta R}{R} = P(a/h)^2 (1-\nu)\pi$$









Bosch BMP085
5 mm x 5 mm x 1.2 mm

K. E. Petersen, "Silicon as a mechanical material," in *Proceedings of the IEEE*, May 1982
A. A. Barlian et al, "Review: Semiconductor Piezoresistance for Microsystems," in *Proceedings of the IEEE*, March 2009

Piezoelectricity

Direct piezoelectric effect arises from the stress-induced charge polarization in a crystal that lacks a center of inversion. Converse piezoelectric effect is when an applied electric field across a piezoelectric material generates strain. Materials - PZT, AlN, GaN, Quartz, PVDF etc. Silicon doesn't show piezoelectricity.

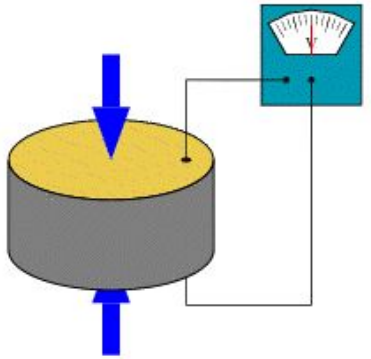
The coupling element is piezoelectric tensor (rank 3)

$$S_{ij} = \sum_{k,\ell} s_{ijk\ell}\, T_{k\ell} + \sum_{k} d^{t}_{ijk}\, E_k,$$

Strain tensor compliance tensor stress tensor piezoelectric tensor electric field

$$D_i = \sum_{j,k} d_{ijk}\, T_{jk} + \sum_{j} \varepsilon_{ij}\, E_j,$$

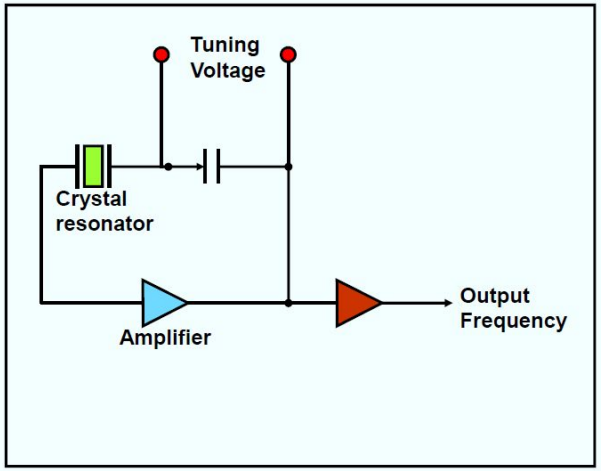
Electric displacement piezoelectric tensor stress tensor permittivity tensor electric field



The vibration could be parallel or perpendicular to electric field depending on the dominant d_{ijk} of the material

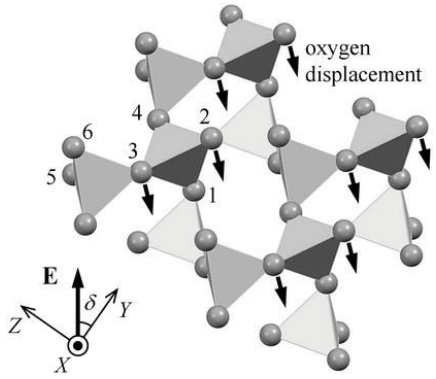
Quartz - Oscillating since 1918

Perhaps the oldest electrical/electronic component still in use

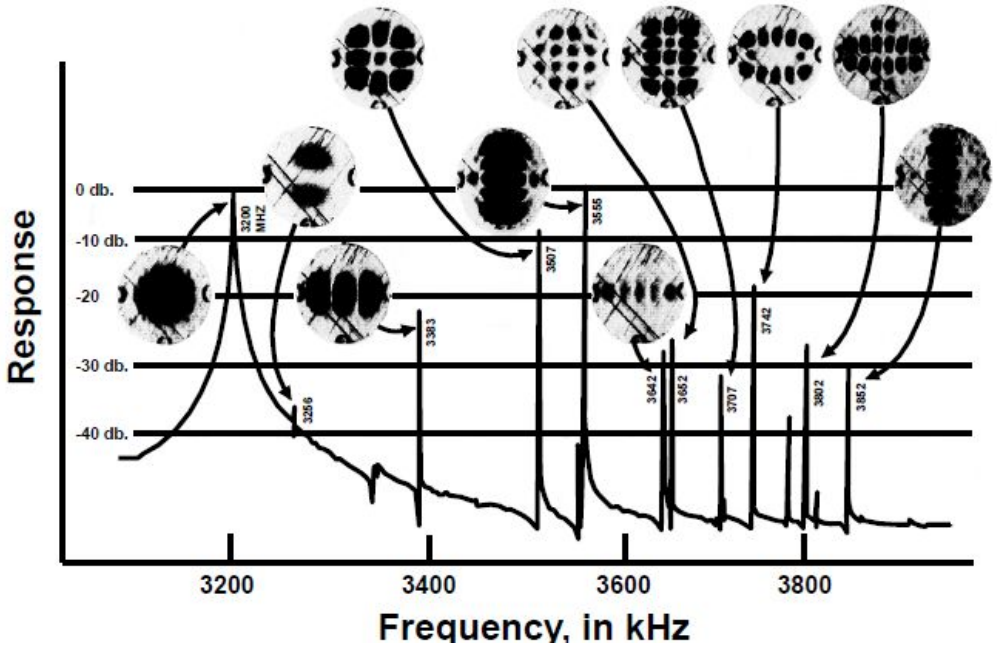


Advantages	Disadvantages
High Q & low phase noise	Process flow not CMOS compatible
Timing Accuracy	Miniaturization bottleneck (Board space)
Stability with Temperature	Difficult to realize > 50 MHz
Low cost	MTTF and Vibration sensitivity

Quartz watches use a 32768 Hz crystal and then divide by 2^{15} to give 1 Hz pulses to a stepping motor for seconds hand



Displacement of O ions in SiO_4 tetrahedra



Contra LCR, resonances, anti-resonances, spurious peaks galore